

Yuvraj Singh Naruka

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EDUCATION

Vellore Institute of Technology, Bhopal

Bachelor of Technology in Computer Science and Engineering , CGPA: 7.47/10

Bhopal, Madhya Pradesh

Oct 2022 – Present

Kendriya Vidyalaya No.2 Jaipur Cantt.

Class XII (CBSE) , Percentage: 89.0%

Jaipur, Rajasthan

Apr 2020 – May 2021

Kendriya Vidyalaya No.2 Jammu Cantt.

Class X (CBSE) , Percentage: 90.2%

Jammu, Jammu and Kashmir

Apr 2018 – May 2019

TECHNICAL SKILLS

Programming Languages: Java, Python, SQL

Web Development: HTML, CSS, JavaScript

Tools: GitHub, Canva

TECHNICAL PROJECTS

Pneumonia Detection | *Python, Keras, TensorFlow*

2025

- Led a team of 5 members to develop a Convolutional Neural Network (CNN) using Python, Keras, and TensorFlow to classify pneumonia from chest X-ray images, achieving 91.1% accuracy on test set.
- Processed and augmented a dataset of 5,000+ Chest X-ray images, performing resizing, normalization, and image transformations to improve model generalization.
- Utilized deep learning techniques including Batch Normalization, Dropout, and learning rate scheduling to optimize model performance, with the final model demonstrating high sensitivity and specificity.

Heart Failure Prediction | *Python, Scikit-learn*

2024

- Led a team of 5 members to develop machine learning models (Random Forest, Logistic Regression, SVM) using Python and Scikit-learn to predict cardiovascular risk, achieving 89.75% accuracy.
- Processed and analyzed 1,000+ patient records, performing data preprocessing and cleaning to enhance model reliability by 30%.
- Utilized feature selection techniques and cross-validation to optimize model performance, with Random Forest delivering the highest accuracy.

Loan Creditworthiness Prediction | *Python, Scikit-learn*

2024

- Developed machine learning model (Random Forest Algorithm) using Python and Scikit-learn to predict loan creditworthiness, achieving optimal performance through RandomizedSearchCV with 5-fold cross-validation.
- Processed and analyzed loan application data with 18 features, implementing data preprocessing pipelines with standardization and one-hot encoding to enhance model reliability.
- Utilized class weighting (10:1 ratio) and probability scoring to optimize model performance, with Random Forest delivering superior results in predicting credit risk through F1-score optimization.

CERTIFICATIONS

Oracle Cloud Infrastructure Foundations Associate - *Oracle University*

Oracle Cloud Infrastructure AI Foundations Associate - *Oracle University*

EXTRACURRICULAR ACTIVITIES

Corresponding Author — *Infectious Diseases Now Journal*

2025

- Published editorial "Balancing AI accuracy and medical guidance: Evaluating ChatGPT's role in UTI-related queries" in peer-reviewed medical journal (Volume 55, Issue 3).
- Contributed to academic discourse on AI applications in healthcare and infectious disease research, addressing ethical implications and limitations of AI-driven medical consultations.

AIEM Club — *Member*

2024

KVS Jammu Regional Basketball Team — *Vice Captain*

2019